

Totally Integrated Automation Portal

Main [OB1]

Main Properties

General

Name	Main	Number	1	Type	OB	Language	LAD
Numbering	Automatic						

Information

Title	"SmartLine 360 — Main Program Sweep"	Author	Youness Tebbaq	Comment	"Main program sweep. Calls the lab function only in the first cycle, then runs the three stations in sequence: Station 1 (Infeed & Color Sorting) → Station 2 (Assembly) → Station 3 (Packaging & Palletizing). Every station is a Function Call (FC) so that each one can be compiled and tested independently."		
Version	0.1	User-defined ID					

Name	Data type	Default value	Comment
▼ Input			
Initial_Call	Bool		Initial call of this OB
Remanence	Bool		=True, if remanent data are available
Temp			
Constant			

Network 1: Factory I/O Communication & System Initialization Bridge

Purpose:

This network executes the library block FC0000 (MHU-PLC-Lab-Function-S71200), which manages the low-level communication drivers and data exchange between TIA Portal and the Factory I/O simulation software.

Behavior:

Executes during the first scan cycle only (Initial_Call).

Establishes the I/O driver handshake, initializes global data block starting values, and syncs system timing.

Note: Do not delete or disable this network while running the system in simulation with Factory I/O, as it will break the PLC-to-Factory I/O communication link.

%FC9000

"MHJ-PLC-Lab-Function-S71200"

EN

ENO

Network 2: Station 1 — Infeed & Color Sorting

Station 1 (Infeed & Color Sorting). Emitter spawns blue/green lids into the line; the vision sensor detects the color and the pusher ejects the blue lids onto a divert lane after a short timer delay. FC1 runs continuously every PLC cycle calls it.

%FC1

"Station1"

EN

ENO

Network 3: Station 2 — Assembly

Station 2 (Assembly). Two emitters feed a Product Base and a Product Lid(from station 1) onto separate short belts. Stop blades hold each part in place, then the Two-Axis Pick & Place picks the lid, moves it to the base and releases it to form the Final Product. FC2 runs continuously.

%FC2

"Station2"

EN

ENO

Network 4: Station 3 — Packaging & Palletizing

Station 3 (Packaging & Palletizing). Final products arrive via a roller conveyor and are pushed into an elevator one row at a time. The elevator raises, a clamp/plate sequence loads the pallet, and the elevator lowers again. When a pallet is full it exits onto an outfeed roller conveyor. FC3 runs continuously.

%FC3

"Station3"

EN

ENO